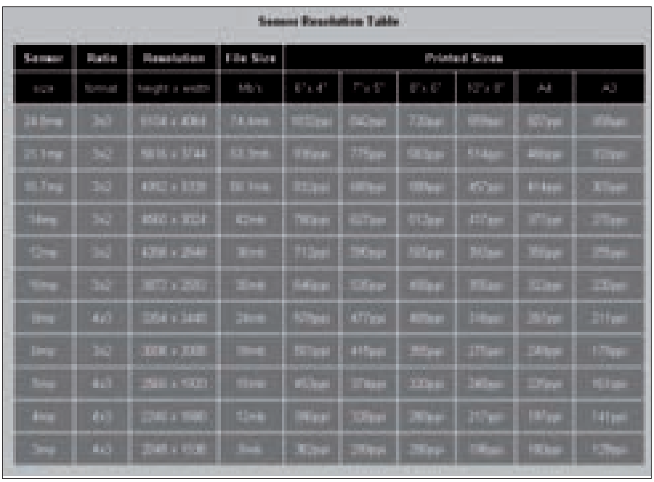
Most P&S cameras have a crop factor of 1.3 to 2.0. This is what marks the difference between 'number of sensor pixels' and 'effective pixels'. Some of the cameras lose effective pixels because the camera's lens can't cover the entire sensor area and so what you get is actually a picture that's been cropped all around. At the same time, full frame sensors like the Canon 1D, sometimes lead to a visible fringing – a lateral chromatic aberration – which you'll need to crop with editing software after the fact. But all said and done, the full frame sensor gives you a wide-angle of view (therefore capturing more of the scene) and a better quality image.

14.5 Sensor size and image resolution

Cameras with smaller sensors use shorter focal length lenses to get the same angular coverage as cameras with larger sensors do with longer focal length lenses. So what's the big deal about sensor size, then? The size of the sensor affects many factors in the image and the functions of the camera and its body. Basically, sensors are available in three different sizes : four thirds, APS and Full Film Format.



Sensor Resolution Table

Sensor size	Ratio format	Resolution length x width	File Size Mb's	Printed Sizes 6" x 4"	7" x 5"	8" x 6"	10" x 8"	A4	A3
[illegible]	3x2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	3x2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	3x2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	3x2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	3x2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	3x2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	4x3	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	3x2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	4x3	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	4x3	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	4x3	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Making sense of sensor resolution